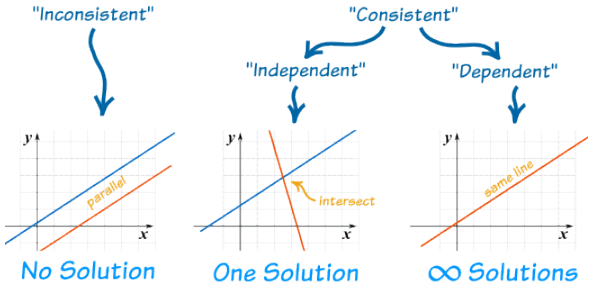
Intersection of Two Lines

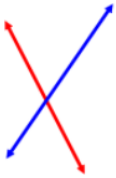
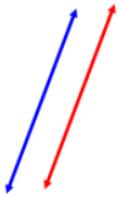
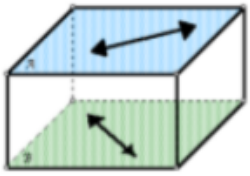
Recall: In R^2, solution(s) to a linear system, graphically, are _____.



A linear system is said to be _____ if it has at least one solution.

Otherwise, the system is said to be _____.

In R^3, two lines can intersect in the following ways:

One Solution	Infinite Solutions	No Solution	No Solution
			
Intersecting Lines	Coincident Lines	Parallel and Distinct Lines	Skew Lines *(Planes containing the lines are parallel, but the lines themselves are not)*

To solve a system algebraically, we can use the methods of substitution/elimination.

In R^2, given Cartesian equations ($Ax + By + C = 0$), we can simply use _____ or _____.

If we were given vector equations…

Example 1: Determine the point of intersection between the two lines:

$$\vec{r}_1 = (-1,1,0) + t(3,4,-2),\ t \in \Re$$
$$\vec{r}_2 = (-1,0,-7) + s(2,3,1),\ s \in \Re$$

The other option to solve is to write one equation in parametric form and sub into the other either in symmetric form (R^3) or Cartesian form (R^2)

Determine the point of intersection: $\vec{r}_1 = (-1,1,0) + t(3,4,-2),\ t \in \Re$

$\vec{r}_2 = (-1,0,-7) + s(2,3,1),\ s \in \Re$

Example 2: Determine the point of intersection between the two lines:

$\vec{r}_1 = (1,2,3) + t(1,-1,2),\ t \in \Re$

$\vec{r}_2 = (3,1,4) + s(4,1,3),\ s \in \Re$

Example 3: Determine the point of intersection between the two lines:

$\vec{r}_1 = (-3,4) + t(2,-1),\ t \in \Re$

$\vec{r}_2 = (-7,1) + s(-2,1),\ s \in \Re$